

Pinned group summary: $SO(n=4, q=1)$

Pinning-Sympy automated output

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1 Basic group attributes

Group	$\mathrm{SO}(n=4, q=1)$
Matrix size	4×4
Rank	1
Defining equations	<i>DefiningEquationPlaceholder</i>

1.1 Symmetric bilinear form

Dimension	4
Witt index	1
Primitive element	N/A
Epsilon	N/A

Matrix:
$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & c_0 & 0 \\ 0 & 0 & 0 & c_1 \end{bmatrix}$$

1.2 Generic Lie algebra element

$$\begin{bmatrix} x_{00} & 0 & -c_0 x_{21} & -c_1 x_{31} \\ 0 & -x_{00} & -c_0 x_{20} & -c_1 x_{30} \\ x_{20} & x_{21} & 0 & x_{23} \\ x_{30} & x_{31} & -\frac{c_0 x_{23}}{c_1} & 0 \end{bmatrix}$$

1.3 Generic torus element

$$\begin{bmatrix} t_0 & 0 & 0 & 0 \\ 0 & \frac{1}{t_0} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

1.4 Trivial characters

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

2 Root system

Summary Properties

Dynkin type	A1
Irreducible	True
Reduced	True
Simply laced	True
Number of roots	2

2.1 Dynkin diagram



2.2 Cartan matrix

[2]

2.3 Roots and coroots

Root	Coroot	Simple	Sign	Multipliable
(−1, 0, 0, 0)	(−2, 2, 0, 0)	No	−	No
(1, 0, 0, 0)	(2, −2, 0, 0)	Yes	+	No

2.4 Linear combinations of roots

Definition 2.1. For two roots α, β in a root system Φ , the set of positive integral linear combinations of α, β that are also roots is denoted $(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$

No pairs of roots generate positive integer linear combinations.

3 Root spaces

Root	Dimension
(−1, 0, 0, 0)	2
(1, 0, 0, 0)	2

RootSpaceTablePlaceholder

4 Root subgroups

RootSubgroupTablePlaceholder
HomomorphismDefectTablePlaceholder
HomomorphismDefectIdentitiesPlaceholder

5 Commutators

CommutatorCoefficientTablePlaceholder
CommutatorIdentitiesPlaceholder

6 Weyl group

WeylGroupPlaceholder
WeylConjugationCoefficientPlaceholder
WeylConjugationIdentitiesPlaceholder

7 Coroot torus elements (h_α)

CorootTorusElementPlaceholder